

A graded catalytic-protective layer for an efficient and stable water-splitting photocathode

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Achieving solar-to-hydrogen efficiencies above 15% is key for the commercial success of photoelectrochemical water-splitting devices. While tandem cells can reach those efficiencies, increasing the catalytic activity and long-term stability remains a significant challenge. Here we show that annealing a bilayer of amorphous titanium dioxide (TiO_x) and molybdenum sulfide (MoS_x) deposited onto GaInP₂ results in a photocathode with high catalytic activity (current density of 11 mA cm⁻² at 0 V versus the reversible hydrogen electrode under 1 sun illumination) and stability (retention of 80% of initial photocurrent density over a 20 h durability test) for the hydrogen evolution reaction. Microscopy and spectroscopy reveal that annealing results in a graded MoS_x/MoO_x/TiO₂ layer that retains much of the high catalytic activity of amorphous MoS_x but with stability similar to crystalline MoS₂. Our findings demonstrate the potential of utilizing a hybridized, heterogeneous surface layer as a cost-effective catalytic and protective interface for solar hydrogen production.

Hydrogen, utilized in fuel cells to power electric motors or burned in internal combustion engines, is an environmentally friendly energy carrier with the potential to reduce our dependence on fossil fuels. However, the production of hydrogen by the traditional gasification of coal and oil and by steam-methane reforming produces large amounts of carbon dioxide, which has implications for climate change^{1,2}. An alternative long-term, sustainable pathway to hydrogen production is a photoelectrochemical (PEC) cell that absorbs sunlight and converts this energy into hydrogen and oxygen via the dissociation of water molecules^{3,4}. Oxide semiconductor materials, such as Fe₂O₃, WO₃, SrTiO₃ and TiO₂, have been studied for many years for PEC water splitting^{5,6}. However, the slow charge transport kinetics and/or large bandgaps that typically define these oxide semiconductors result in very low energy conversion efficiencies. In contrast, conventional photovoltaic semiconductors, such as GaInP₂ (refs 7–9), offer excellent transport properties and smaller bandgaps than oxides, but are susceptible to corrosion during the water-splitting process¹⁰. To realize and commercialize future solar hydrogen concepts based on PEC devices, durability of tens of thousands of hours and a device cost of hundreds of dollars per square metre must be achieved¹¹.

While certainly a high-cost material, the choice of GaInP₂ for this study stems from it being the only wide-bandgap semiconductor that could be paired with a lower-bandgap material such as silicon or GaAs in a tandem water-splitting device and match the efficiencies of those semiconductors. In the tandem configuration shown in ref. 7, the GaInP₂ is exposed to the solution, while the lower-bandgap material (in the case here it is GaAs, but silicon is also possible) is protected from the electrolyte. Thus, we feel it is important to explore possible surface stabilization and catalytic approaches for this material.

Recently, low-cost TiO₂ coating strategies have been shown to slow photodegradation of the light-absorbing semiconductor in PEC devices^{12–15}. Coupling such protective coatings with a non-precious metal hydrogen evolution reaction (HER) catalyst offers the most promising alternative to expensive Pt and Ru HER catalysts. For instance, Ni alloys have shown high catalytic activity for the HER reaction, but are thermodynamically limited because they are unstable in acidic solution under which proton reduction is frequently studied¹⁶. On the other hand, crystalline molybdenum sulfide (c-MoS₂) is another inexpensive and promising HER catalyst that offers acid tolerance¹⁷, but its usefulness is limited by its layered structure that hinders both its conductivity (electron transfer is preferred along its layers, orthogonal to the necessary electron transfer direction from semiconductor to electrolyte) and activity (minimal exposed active edge sites)¹⁸. In contrast to c-MoS₂, amorphous MoS_x (a-MoS_x) has many defect sites and coordinately unsaturated S atoms that provide significantly higher activity for the HER. However, its poor crystallinity leads to relatively high solubility and, thus, a-MoS_x exhibits limited electrochemical stability under the strongly acidic or alkaline conditions during sustained photoelectrolysis^{19–21}. For instance, deposition of a-MoS_x/amorphous titania (a-TiO_x) a Cu₂O photocathode improves stability in weakly acidic (pH 4) and basic (pH 9) media for up to 10 h, but the electrode quickly corrodes in strong acid (pH 1) (ref. 20).

Although the catalyst/TiO₂ construction strategy has been widely applied to protect photoelectrodes, only few have focused on in-depth understanding of the structure and chemistry of the interface²². Here, we show that annealing a bilayer of a-MoS_x/TiO_x, applied to a GaInP₂ photoelectrode, results in a graded g-MoS_x/MoO_yS_z/MoO_x/c-TiO₂ interfacial layer, endowing

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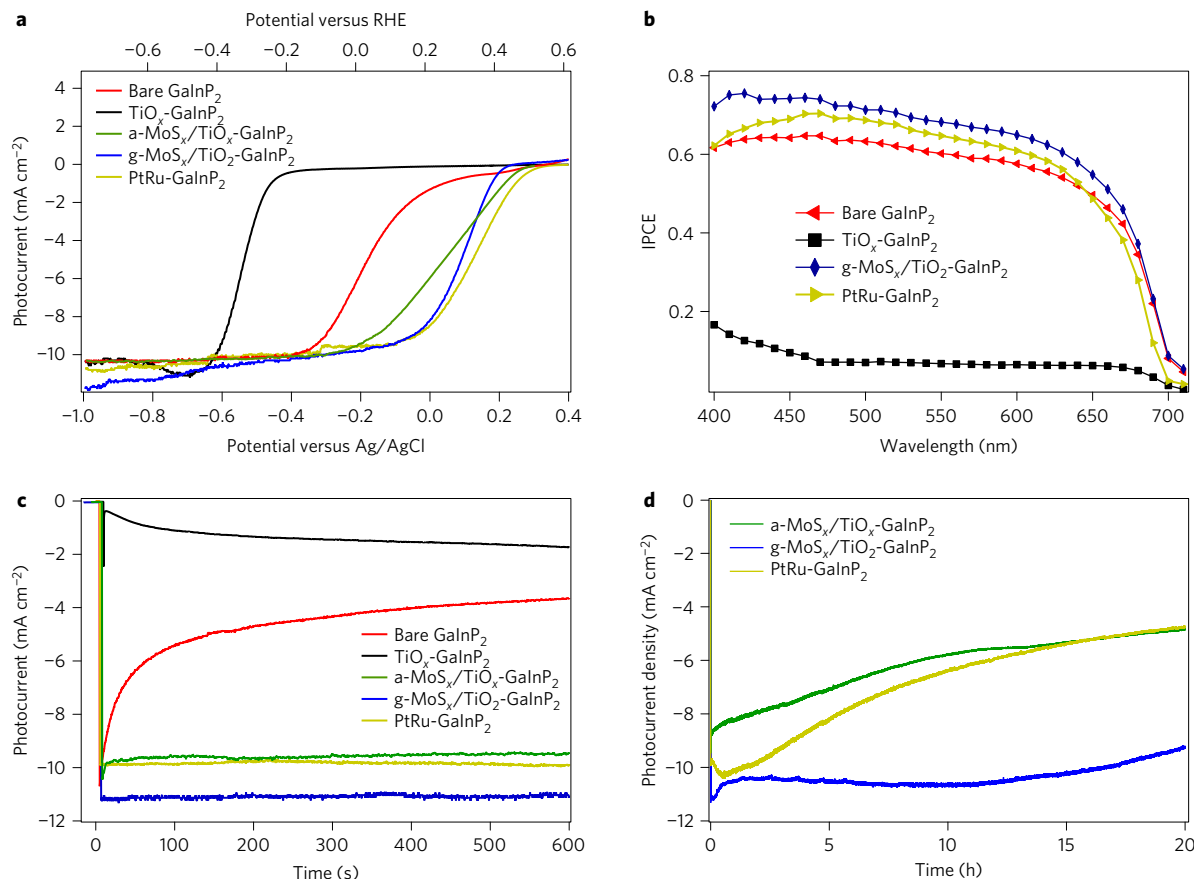


Figure 1 | Photoelectrochemical measurements of GaInP₂-based photocathodes for hydrogen evolution. **a**, Photocurrent density-potential (J - V) curves (scan rate is 20 mV s⁻¹) in 0.5 M H₂SO₄ under 1 sun illumination. **b**, Incident photon-to-current conversion efficiency (IPCE) of various GaInP₂ electrodes in 0.5 M H₂SO₄ at 0 V versus RHE. **c**, 10 min photocurrent density-time (J - t) plots of different GaInP₂ electrodes held at 0 V versus RHE in 0.5 M H₂SO₄ under 1 sun illumination. **d**, 20 h photocurrent density-time (J - t) plots of different GaInP₂ electrodes held at 0 V versus RHE in 0.5 M H₂SO₄ under 1 sun illumination.

the photocathode with PEC properties superior to those of a GaInP₂ electrode coupled with a PtRu alloy HER catalyst. We study the changes in the catalyst morphology, composition and activity prior to and after annealing and show that the high activity and stability arises from both the generation of a number of Mo-containing phases including g-MoS_x and MoO_x, and crystalline TiO₂ (c-TiO₂) at the surface of the electrode. Finally, we discuss how these observations suggest a general design principle for a hybrid catalyst-protection layer architecture.

Surface modification of GaInP₂

p-GaInP₂ wafers were grown at the National Renewable Energy Laboratory using metal-organic vapour-phase epitaxy as described in the Methods^{7,23}. Amorphous titanium dioxide (a-TiO_x) layers were deposited by atomic layer deposition onto bare p-GaInP₂ electrodes to create a-TiO_x-GaInP₂ samples. Amorphous MoS_x was deposited onto a fresh a-TiO_x-GaInP₂ sample using electrochemical cathodic deposition, as described in the Methods^{19,24,25}, to form a-MoS_x/a-TiO_x-GaInP₂, which was annealed at high temperature (450 °C) to give g-MoS_x/MoO_x/c-TiO₂-GaInP₂. For brevity, we refer to g-MoS_x/MoO_x/c-TiO₂-GaInP₂ as g-MoS_x/TiO₂-GaInP₂ for the rest of the ensuing text. For benchmarking, a PtRu catalyst was deposited onto bare GaInP₂ via sputtering as described previously to yield PtRu-GaInP₂ (ref. 26). The HER catalytic activities of these electrodes were measured by linear sweep voltammetry scanning in the cathodic direction in Ar-purged 0.5 M H₂SO₄ solutions (pH 0.3) under 1 sun conditions (for all electrochemical experiments). All electrodes under dark conditions show no measurable HER activity

in the potential window. See the Methods for a detailed description of all materials and methods used in this study.

Photoelectrochemical profile

Representative photocurrent density-potential (J - V) plots are shown in Fig. 1a for bare GaInP₂, TiO_x-GaInP₂, a-MoS_x/a-TiO_x-GaInP₂, g-MoS_x/c-TiO₂-GaInP₂ and PtRu-GaInP₂ electrodes. As can be seen from these data, the a-TiO_x-GaInP₂ electrode (black curve) reached a saturated photocurrent density (~10 mA cm⁻²) at about 500 mV overpotential versus the reversible hydrogen electrode (RHE) (see about 150 mV for bare GaInP₂, red curve) consistent with the well-known poor HER activity of a-TiO_x. Adding a layer of a-MoS_x caused the onset potential to shift positively and the saturated photocurrent density to be reached at near-zero overpotential (green curve), albeit with a low fill factor. Converting the a-MoS_x/a-TiO_x-GaInP₂ to g-MoS_x/c-TiO₂-GaInP₂ (blue curve) via annealing resulted in a substantially improved fill factor, probably due to improved electron transport through the now mostly crystalline interfacial layer; however, the onset potential decreased slightly, presumably owing to the greater number of active sites in the a-MoS_x layer in the unannealed sample. A PtRu catalyst-decorated electrode (PtRu-GaInP₂, yellow curve), which is known to exhibit HER catalytic activity superior to Pt or Ru alone, is employed here as a benchmark system^{26,27}. As can be seen from Fig. 1a, the catalytic activity of the annealed g-MoS_x/c-TiO₂-GaInP₂ sample is comparable to PtRu-GaInP₂, as evidenced by similar 0.2 V underpotentials at the saturated photocurrent density, albeit with a ~100 mV more positive

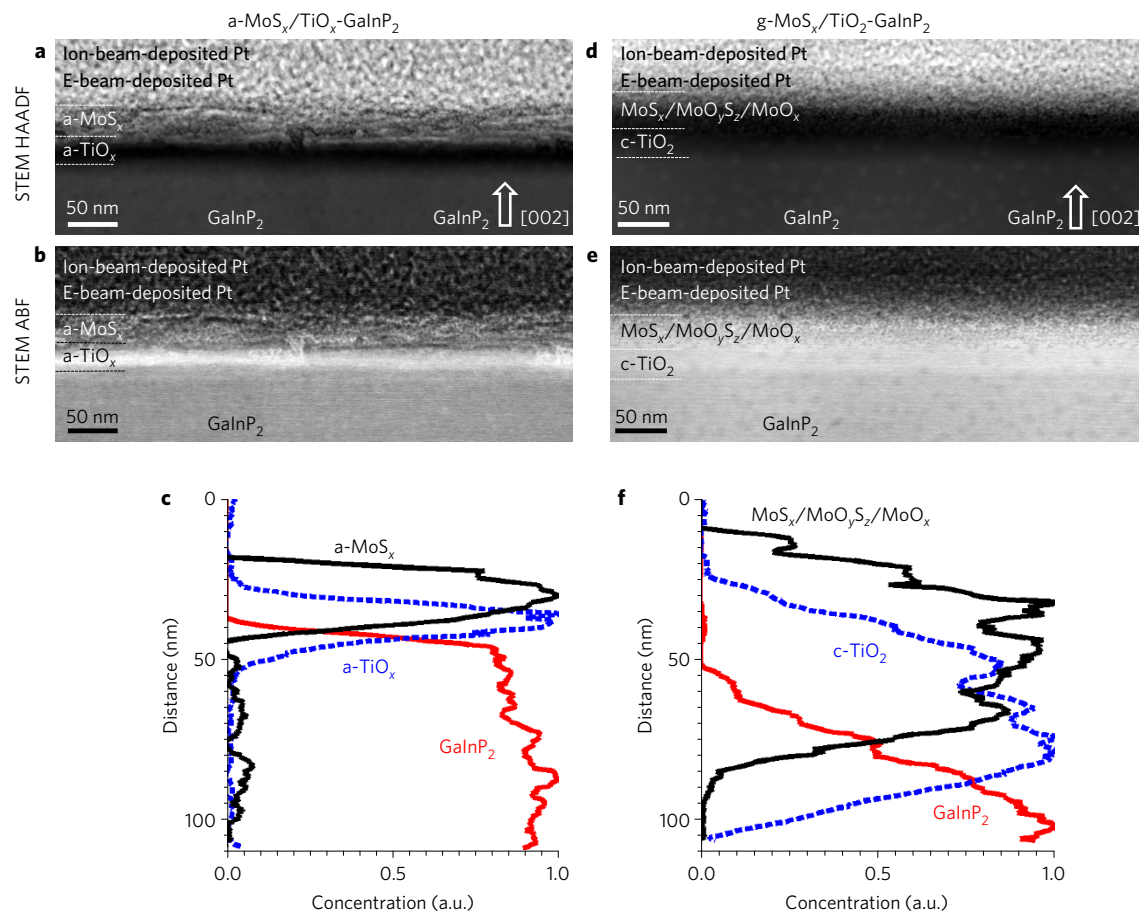


Figure 2 | Structural and chemical profiling of the a-MoS_x/TiO_x-GaInP₂ and g-MoS_x/TiO₂-GaInP₂ electrodes. a,b, Cross-sectional high-angle annular dark-field (HAADF) (a) and annular bright-field (ABF) (b) STEM image of a-MoS_x/TiO_x-GaInP₂ before annealing. **c**, STEM EDS line profiling coupled with spectral component matching showing the individual layer thickness (a-MoS_x, black; TiO_x, blue; GaInP₂, red) and interface abruptness before annealing. **d,e**, Cross-sectional HAADF (d) and ABF (e) STEM image of g-MoS_x/TiO₂-GaInP₂ after annealing. **f**, STEM EDS line profiling coupled with spectral component matching showing the individual layer thickness (MoS_x/MoO_yS₂/MoO_x, black; TiO₂, blue; GaInP₂, red) and evidence of inter-diffusion between layers after annealing. Note, given the samples were lifted out using a focus ion instrument, electron and ion beam Pt was deposited on the surface to protect the sample during thinning to electron transparency and each of these Pt layers is indicated.

onset potential. This result is consistent with the onset potential difference of 145 mV between Pt and MoS_x catalysts on a Ti-n⁺p-Si photoelectrode²⁸. Finally, the photocurrent density values were consistent with IPCE measurements (Fig. 1b), which displayed a high conversion efficiency (up to 75% across the visible range) from the annealed MoS₂/TiO₂-GaInP₂ electrode, ~10% higher than the PtRu-modified electrode, and ~20% higher than the bare GaInP₂.

Material stability and hydrogen production

Photocurrent density versus time was monitored at 0 V versus RHE to evaluate the short-term (10 min) durability of the electrodes. The cathodic current density of bare GaInP₂ exhibited rapid decay owing to photocorrosion as is well known¹⁰. The a-TiO_x-GaInP₂ sample showed an apparent increase in photocurrent density to 1–2 mA cm⁻² over this time frame, but this is probably also attributed to photocorrosion due to the limited thermodynamic stability of TiO_x in acidic media²⁹. In contrast, all samples with catalytically active layers (a-MoS_x/a-TiO_x-GaInP₂, g-MoS_x/c-TiO₂-GaInP₂ and PtRu-GaInP₂) exhibited little photocurrent decay during the 10 min photoelectrolysis experiment (Fig. 1c). The initial charging spike from the a-MoS_x/a-TiO_x-GaInP₂ sample may be due to the capacitance change and does not reflect the real current³⁰.

Longer-term stability of these three samples was monitored for 20 h under sustained photoelectrolysis at 0 V versus RHE. As can be seen from Fig. 1d, the photocurrent density of the unannealed

electrode a-MoS_x/a-TiO_x-GaInP₂ decreased significantly, from 10 mA cm⁻² to 5 mA cm⁻², after 20 h. This drop in photocurrent density is apparent both without any a-TiO_x layer (MoS_x-GaInP₂, Supplementary Fig. 1) or with annealed c-TiO₂ prior to electrodeposition of a-MoS_x (a-MoS_x/c-TiO₂-GaInP₂, Supplementary Fig. 2), suggesting that the instability results from the a-MoS_x layer consistent with prior reports^{18,31}. Interestingly, the majority of the decrease in photocurrent density from the a-MoS_x/c-TiO₂-GaInP₂ sample occurred within the first 10 h, with relatively stable performance observed from 10 to 20 h (Supplementary Fig. 1). This result was not explored in further detail in this work but may motivate future experiments of mixed interfacial layers. In contrast to unannealed a-MoS_x/a-TiO_x-GaInP₂, the high photocurrent density produced by g-MoS_x/c-TiO₂-GaInP₂ diminished by less than 20% over 20 h, which is also substantially better durability than that observed for the PtRu-GaInP₂ sample (Fig. 1d). The instability of the PtRu-GaInP₂ sample appears to be related to the incomplete coverage of the PtRu alloy over the GaInP₂ surface, which forms small (1–10 nm) particles on GaInP₂ as evidenced by atomic contrast scanning transmission electron microscopy (STEM) imaging (Supplementary Fig. 3). Hydrogen production was confirmed by gas chromatography, which showed 94% Faradaic efficiency over the initial 6 h of photoelectrolysis (Supplementary Fig. 4). After 20 h, a turnover number (TON) of 367,000 and turnover frequency (TOF) of 5.1 s⁻¹ was calculated based on analysis

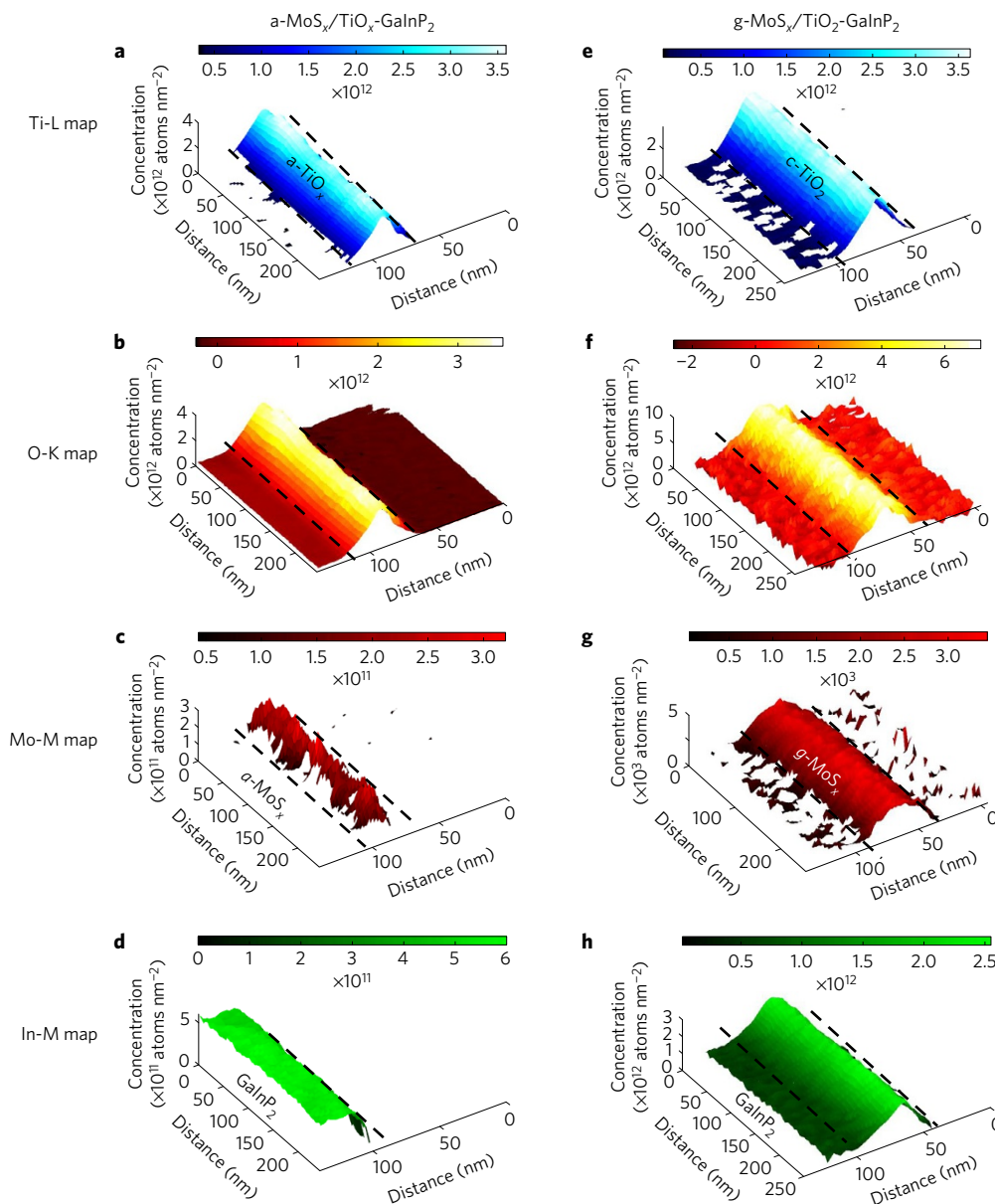


Figure 3 | Chemical profiling of the non-annealed (a-MoS_x/TiO_x-GaInP₂) and annealed (g-MoS_x/TiO₂-GaInP₂) electrodes. The mapping shows the individual layer thickness associated with the Ti, O, Mo and In elements. **a–h**, Integrated line plots and chemical analyses are used to show the layer topology associated with the as-deposited sample MoS_x/TiO_x-GaInP₂ (**a–d**) and annealed g-MoS_x/TiO₂-GaInP₂ (**e–h**) where each layer is delineated by the dashed lines based on the complementary STEM image. Each of these 3D surface plots is further reported in calculated atomic densities, where each plot is intensity colour scaled to the total number of atoms per square nanometre. The as-deposited sample for Ti (**a**), O (**b**), Mo (**c**) and In (**d**) clearly delineates each layer. In the case of the annealed sample, comparing each of the EELS component maps for Ti (**e**), O (**f**), Mo (**g**) and In (**h**) demonstrates that there is a higher degree of intermixing on the order of 5 nm between the annealed TiO₂ and MoS₂ layers.

of the catalyst loading ($10.49 \text{ nmol cm}^{-2}$ or $1.68 \mu\text{g cm}^{-2}$) using inductively coupled plasma mass spectrometry (Supplementary Table 1). Scanning electron micrographs of a-TiO_x-GaInP₂, a-MoS_x/a-TiO_x-GaInP₂ and g-MoS_x/c-TiO₂-GaInP₂ before and after 20 h of electrolysis are shown in Supplementary Fig. 5C and D, respectively. Following electrolysis, there is no discernible change in the MoS₂ particle size, although some unknown material appears to have deposited in a few small patches.

Chemical properties of g-MoS_x/TiO₂-GaInP₂

These data demonstrate that the annealed g-MoS_x/TiO₂-GaInP₂ sample is similarly active for HER at short times and retains even more of its initial activity than PtRu-GaInP₂. We inferred that the high activity and stability of g-MoS_x/c-TiO₂-GaInP₂ stemmed both

from the well-known protection abilities of the intermediate c-TiO₂ layer^{15,28} and some unique features of the annealed g-MoS_x and its interface with c-TiO₂.

We probed the chemical nature of the interface before and after annealing in more detail using TEM, STEM, electron energy-loss spectroscopy (EELS) and X-ray photoelectron spectroscopy (XPS). Figure 2c displays the cross-sectional STEM annular bright-field (ABF) image of an as-deposited a-MoS_x/a-TiO_x-GaInP₂ sample showing relatively well-defined layers of a-MoS_x, a-TiO_x and GaInP₂ owing to uniform surface deposition. STEM-based energy-dispersive X-ray spectroscopy (EDS) line profiling coupled with spectral component matching confirmed the sharp profiles of each layer (Fig. 2e). Spectral component matching was performed by calculating the maximum spectrum from each area of the

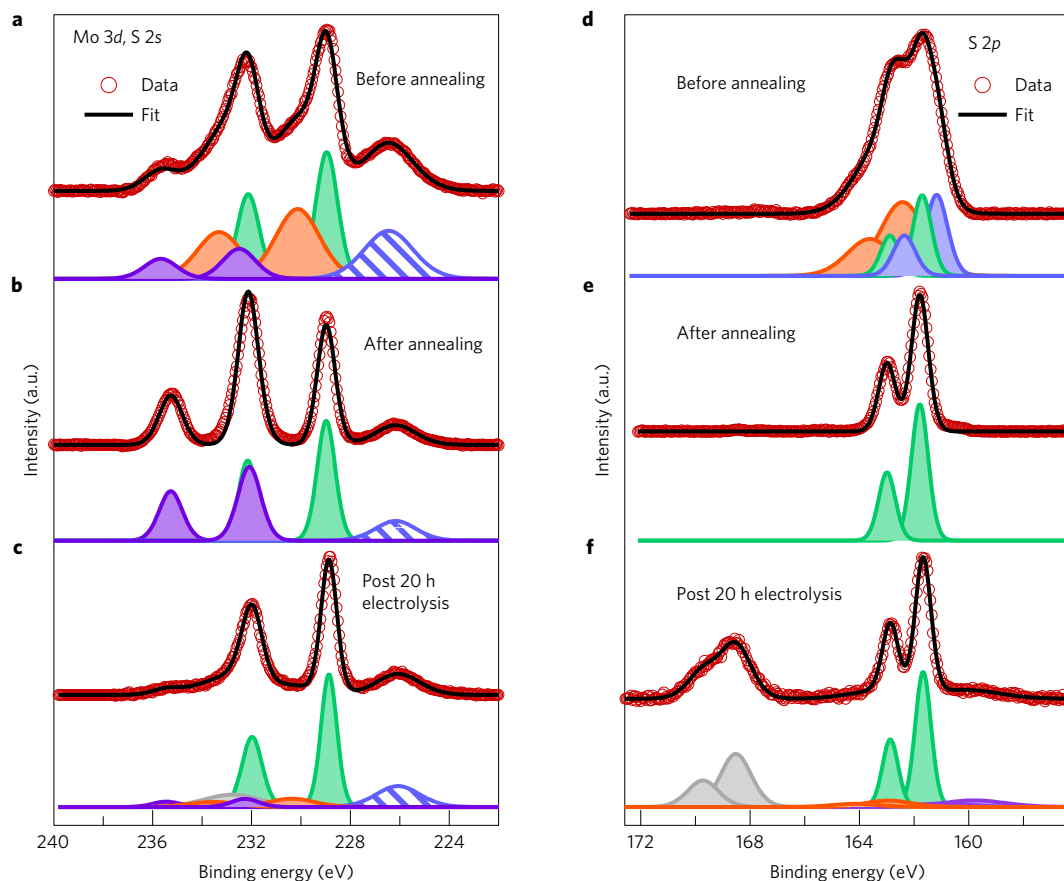


Figure 4 | XPS spectra of the MoS_x/TiO_x-GaInP₂ electrode before and after annealing and before and after electrolysis for 20 h. a–c, XPS of Mo 3d and S 2s regions for a-MoS_x/a-TiO_x-GaInP₂ (a), a-MoS_x/TiO_x-GaInP₂ before annealing (b) and g-MoS_x/c-TiO₂-GaInP₂ (c) after 20 h photoelectrolysis showing experimental data (red dots), fitting envelope (black line), and peak fits corresponding to Mo(IV) from MoS_x (green), Mo(V) from MoO_yS_z (orange), Mo(VI) from MoO_x (purple), various sulfur species (S₂²⁻, S²⁻, and another electron-rich sulfur phase (for example, S²⁻ from residual (NH₄)₂S); hashed blue), and S⁶⁺ from sulfate (grey). **d–f**, XPS of the S 2p region for a-MoS_x/a-TiO_x-GaInP₂ (d), g-MoS_x/c-TiO₂-GaInP₂ (e) and g-MoS_x/c-TiO₂-GaInP₂ (f) after 20 h photoelectrolysis showing experimental data (red dots), fitting envelope (black line), and peak fits corresponding to S⁶⁺ from sulfate (grey), S₂²⁻ (orange), S²⁻ (green), S⁰ (solid blue), and electron-rich sulfur (purple).

sample. This partitioning of the spectral data set allows each component respective of each layer to be qualitatively plotted as an integrated profile for each of the components. In this manner, the individual groupings used and plotted are Ti/O, Mo/S and In/Ga/P to determine layer thickness. The layer thicknesses of a-MoS_x and a-TiO_x were found to be 22–26 nm and 24–30 nm, respectively, in the non-annealed sample. In contrast, the STEM bright-field image of the annealed sample g-MoS_x/c-TiO₂-GaInP₂ exhibited a much less well-defined interface, with EDS line profiling indicating a large degree of interpenetration between g-MoS_x, c-TiO₂ and GaInP₂ (Fig. 2f). STEM-EELS mapping was used to further probe the interface structure and also showed a clear difference in topology between as-deposited a-MoS_x and a-TiO_x (Fig. 3a,c,e,g) compared with the sample following annealing (Fig. 3b,d,f,h). Comparing the topology based on calculated two-dimensional (2D) atomic densities, the as-deposited sample distribution for Ti (Fig. 3a), O (Fig. 3b), Mo (Fig. 3c) and In (Fig. 3d) resolves no significant layer-to-layer overlap. The STEM-EELS maps qualitatively agree with the EDS results, with the Ti, O, Mo and In maps showing, however, the annealed sample contains significantly more overlap between g-MoS_x/c-TiO₂ than between a-MoS_x/a-TiO_x in the as-deposited sample. Specifically, the EELS component maps for g-MoS_x/c-TiO₂ demonstrate a higher degree of intermixing on the order of 5 nm between the annealed TiO₂ and MoS₂ layers based on the reported Ti (Fig. 3e), O (Fig. 3f), Mo (Fig. 3g) and In (Fig. 3h) calculated atomic density plots. We note the EELS fine

structure is not presented due to overlapping between layers and different localized nanoscale chemistry rendering EELS insufficient for accurate valence state determination. Interestingly, the EELS In map additionally shows more In content near the sample surface and less approaching the GaInP₂ surface, suggesting that In also diffuses into the g-MoS_x/c-TiO₂ interfacial layer following annealing.

XPS analysis was used to further probe the chemical composition of the interfacial layer before and after annealing. Our electrodeposition conditions are comparable to the potentiostatic deposition conditions reported previously³², for which XPS analysis revealed a film of ‘amorphous MoS₂’ plus some hydroxide and hydrosulfide. Our XPS data for the as-deposited a-MoS_x/a-TiO_x-GaInP₂ sample (Fig. 4a,b) show chemical environments for Mo and S in this near-surface (7–9 nm) region similar to that identified in the prior report³². Fitting details for the Mo 3d and S 2p XPS data are provided in the Methods. Three Mo 3d doublets and two broad S 2s peaks were combined to fit the Mo 3d spectra. The Mo 3d_{5/2} binding energy of 228.7 eV (green peaks, Fig. 4a) is attributed to Mo(IV) (for example, MoS₂ or MoS_x), and that at 230.5 eV (orange peaks, Fig. 4a) to a Mo(V) phase that previously has been assigned as MoO_yS_z (ref. 33). The doublet with a Mo 3d_{5/2} energy of 232.5 eV (purple peaks, Fig. 4a) is due to Mo(VI), possibly due to formation of MoO₃ (ref. 32). However, no crystalline MoO₃ was detected from the STEM ABF imaging on the annealed sample (Fig. 2), and thus we use MoO_x to represent the Mo(VI) phases detected here. The S 2s peak at 226.0 eV (hashed blue peak, Fig. 4a) indicative of S²⁻

is consistent with the formation of a Mo(IV) phase such as MoS_x (ref. 32). The S 2*p* spectrum (Fig. 4b) was deconstructed into three doublets assigned to S₂^{2−} (162.4 eV) and S^{2−} (161.7 eV) as well as another electron-rich sulfur phase (161.2 eV, for example, S^{2−} from residual precursor ((NH₄)₂S) consistent with the Mo(IV), Mo(V) and Mo(VI) phases found in the Mo 3*d* region.

After annealing, the XPS spectra change dramatically to eliminate the Mo(V) spectral intensity. The Mo 3*d* region for the g-MoS_x/TiO₂-GaInP₂ electrode (Fig. 4c) exhibits just two Mo oxidation states as evidenced by doublets with Mo 3*d*_{3/2} of 228.7 eV (green peaks) and 232.5 eV (purple peaks) indicative of a MoS_x phase similar to crystalline Mo(IV) (MoS₂) and Mo(VI) (MoO₃), respectively, as described above, and the corresponding S 2*p* spectrum (Fig. 4d) stems entirely from S^{2−} species. Interestingly, the relative amount of Mo(VI) increases substantially, which suggests that annealing under inert atmosphere (Ar, 4 h, 450 °C) causes oxidation of the Mo(V) phases to Mo(VI) and complete conversion of undercoordinated and unreacted sulfur. These data are consistent with the STEM-based EDS and EELS chemical mapping data that show an as-deposited a-MoS_x/a-TiO_x interface that becomes graded with formation of g-MoS_x, MoO₃, and TiO₂ inter-mixed phases.

Finally, a sample via XPS after 20 h photoelectrolysis demonstrates a significantly decreased intensity peak from Mo(VI), probably due to MoO₃ dissolution in acid with extended electrolysis, and similar intensity Mo(IV) and S^{2−} peaks, indicative of MoS₂, but also a small amount of Mo(V) (Fig. 4e). Additional peaks were observed attributable to sulfate (grey S 2*p* doublet at 168.5 eV, Fig. 4f; grey S 2*s* peak at 232.8 eV, Fig. 4e) as well as trace of a highly reduced sulfur species (S 2*p* purple doublet at 159.8 eV, Fig. 4f). Dissolution of MoO₃ and re-deposition as MoO₃S₂ or sulfate could be responsible for the observed material on the electrode surface following photoelectrolysis (SEM image, Supplementary Fig. 5D).

Our observations suggest that the stability induced by annealing could result from the formation of a crystalline MoS₂ phase, a widely recognized HER catalyst with excellent catalytic stability^{18,34} and/or the g-MoS_x/c-TiO₂ hybrid structure improving and/or the g-MoS_x/c-TiO₂ hybrid structure improving the interface corrosion resistance in strong acid and improving catalyst adhesion on top of the protection layer. This is in contrast to other molybdenum sulfide catalyst-protected photocathode interfaces without additional metal interface (such as the MoS₂/TiO₂-nanowire Si system showing a 0.3 V (versus RHE) onset potential and 75 min stability in 0.5 M H₂SO₄; ref. 35); the g-MoS_x/c-TiO₂ interface demonstrated a more positive onset potential (0.5 V versus RHE) and stability (20 h) in strong acid solution (Supplementary Table 2). In addition, the hybridization between the molybdenum oxide species and MoS_x might improve MoS_x catalyst's conductivity and diffusion properties. MoO₃, a conductive oxide³⁶, has been extensively applied in polymer solar cells for protecting indium tin oxide from corrosion caused by organic acid³⁷, which could also be important in the improved stability. Moreover, the mixed oxide MoO₃-TiO₂ has been reported to exhibit strong Brønsted acidity³⁸, which might further advance the acid resistance of the annealed electrode.

Conclusions

In conclusion, a g-MoS_x/c-TiO₂-GaInP₂ photocathode for efficient water reduction was demonstrated with just a 20% loss in photocurrent density and a catalyst TON of 367,000 and TOF of 5.1 s^{−1} under sustained photoelectrolysis for 20 h at 0 V versus RHE in strong acid. High durability, exceeding that of a PtRu-sputtered GaInP₂ photoelectrode, is obtained by annealing the as-deposited MoS_x/TiO_x interfacial layers. The structure and chemical composition of the photoelectrode after annealing was analysed by TEM, STEM, EDS, EELS and XPS and revealed a graded catalytic-protection layer comprised of a-MoS_x, MoO₃S₂, MoO₃ and TiO₂. We propose that the high activity and stability of this system results

from the formation g-MoS_x that retains much of the high catalytic activity of amorphous MoS_x owing to the unique graded interface with MoO₃ and TiO₂.

Methods

Chemicals. All reagents, such as [(NH₄)₂MoS₄] (Sigma-Aldrich), were purchased from commercial sources and used as received.

GaInP₂ photoelectrode. The epi layers of GaInP₂ were grown on GaAs substrates by an atmospheric pressure organometallic vapour-phase epitaxial method¹⁰. The thickness of the epitaxial GaInP₂ layer was 0.5–2 μm and the carrier concentration was 2 × 10¹⁷ cm^{−3}.

Electron microscopy. STEM is a method that enables direct imaging and microanalysis of chemistry and structure extending below 1 nm. Using STEM coupled with spectroscopies such as electron energy-loss spectroscopy (EELS) and energy-dispersive X-ray spectroscopy (EDS) couples the high spatial resolution of the STEM with the chemical certainty to detect elements within 2–3 at% and less, depending on the element and its environment. To perform this level of characterization, however, samples for STEM analysis need to be electron transparent. This requirement means the sample needs to be thinned to less than 100 nm to be imaged. In the case of studying MoS_x/TiO_x/GaInP₂ and MoS₂/TiO₂/InGaP₂ layers, both the raw and annealed samples were lifted out, mounted on copper OmniProbe grids, and further thinned to electron transparency (<80 nm) inside a focused-ion beam instrument. Following this, a final polish using a 600 eV argon beam at ±10° was performed to remove excess damage introduced from the high-energy (5 kV) Ga⁺ ions used during the final stage of the focused-ion beam procedure. Both elastically scattered electrons and inelastic electrons were collected to form both simultaneous bright-field and annular dark-field atomic contrast STEM images with a probe of approximately 1.63 Å at an operating acceleration voltage of 200 kV on a FEI Tecnai F20 Ultra Twin Field Emitting Gun (FEG) STEM located at the National Renewable Energy Laboratory (NREL). Under these conditions, we performed imaging and EDS with an energy dispersion of 10 eV per channel with a 50 μs time constant and 3 s dwell time. In the EDS analysis, a field of view measuring 250 nm × 150 nm with a subnanometre pixel size was integrated for tracking the chemical profile between these samples (only up to 110 nm is shown in the chemical profiles in Fig. 2). To form the EDS line profiles shown we acquired the Ga-K, In-L, P-K, Mo-K, S-K, Ti-K and O-K X-ray emission lines. The EDS line profiles based on these spectra were then compiled and processed as components matching GaInP₂, TiO₂ and MoS_x. Final integrated profiles were also processed by horizontal summing; deviations were calculated on the basis of statistics of standard deviations along defined horizontal line profiles. EELS experiments used an aberration-corrected FEI Titan G2 S (operated at 300 kV) equipped with a Gatan Quantum GIF with high-resolution EELS capabilities, allowing for the use of an electron beam with better than 980 meV energy resolution based on the full-width at half-maximum of the zero-loss energy peak. Drift-corrected 2D spectral images were collected parallel and perpendicular to the layers. Based on the Mo-K/L, S-K, Ti-L and O-K ionization edges, the electron energy-loss near-edge fine structure was quantified on the basis of fitted known Hartree-Slater partial cross-sections at 300 kV with a 26 mrad semi-collection angle using the Gatan Digital Micrograph software. For the TEM spectroscopy in Fig. 2, the contrast is indeed fairly similar for each of the layers. For better identifying each layer, the imaging modalities annular bright-field and high-angle annular dark-field STEM are added.

A plan-view TEM sample of a PtRu-GaInP₂ electrode was prepared by mechanically polishing the sample from the substrate side to about 10 μm thickness and then final thinning to electron transparency from the substrate side in an Ar⁺ ion mill. The conditions used were: ion energy 3.5 kV; 13° incidence angle; the sample rotated and cooled using a liquid nitrogen cold stage; and using one ion gun with the GaInP₂ surface protected from re-deposition of ion-sputtered material by a glass coverslip. The thinned sample was then examined by STEM high-angle annular dark field using the FEI Tecnai F20 Ultra Twin FEG STEM at NREL operated at 200 kV. Bright-field TEM images were obtained using an FEI Tecnai Super Twin TEM at NREL operated at 300 kV.

Inductively coupled plasma mass spectrometry. The inductively coupled plasma mass spectrometry (ICP-MS) experiment was conducted on a Thermo Scientific ICAP Q instrument with a CETAC ASX-520 auto sampler. Molybdenum, platinum and ruthenium standards were prepared from Inorganic Ventures MS Mo-10 ppm, Pt-10 ppm and Ru-10 ppm, respectively. The samples were run in standard (STD) mode with Collision Cell Technology (CCT). The ICP-MS samples were prepared by immersing a GaInP₂-TiO₂-MoS_x or GaInP₂-Pt-Ru electrode into aqua regia digestion solution (0.58 ml) overnight and diluting it to 30 ml for analysis.

Photoelectrochemical measurement. Linear sweep voltammetry data were collected using an EG&G Princeton Applied Research VersaStat II potentiostat with a 300 W Xe-arc lamp (Newport). Intensity of light was calibrated by a using GaInP₂ photodiode ($E_g = 1.82$ eV) whose short-circuit current was measured under AM1.5G (NREL). Note that slight variations in sample position and light intensity measurement might lead to illumination intensities with an at least 10% error as shown by the limited photocurrent differences⁹. Photocathodic durability electrolysis experiments were performed in an H-cell incorporating a salt bridge isolated by a glass frit in between the electrolysis compartments. For electrolysis lasting for 20 h, a steady d.c.-powered 250 W tungsten-halogen (Oriel model 66183) lamp with a water filter blocking infrared irradiation was used as the light source, a GaInP₂ photoelectrode was placed in the same compartment with the Ag/AgCl reference and a Pt counter electrode in the auxiliary compartment. The headspace of the cell was continuously purged with Ar. A gas-tight electrochemical set-up was applied to collect and measure the increased volume caused by H₂ production with a volumetric pipette. The collected gas was further measured by gas chromatography (Shimadzu GC-2010 Plus) with a CarboxenTM 1010 PLOT column and a thermal conductivity detector. The gas product was further confirmed to be H₂.

Electrode fabrication and deposition of a-MoS_x films. Ohmic contacts were made by sputtering gold layers (100 nm thick) onto the opposing sides of the sample and attaching copper wires through silver paste (PELCO colloid silver). The assembly was then covered by a glass and sealed with insulating epoxy (Loctite 9462 Hysol) at room temperature overnight. Further, an acid-resistant epoxy (Loctite E-120 HP), cured at room temperature overnight, was applied to prevent the epoxy layer from etching by the acid solution.

Amorphous molybdenum sulfide thin films were prepared by reductive electrodeposition at -0.3 V versus Ag/AgCl from an aqueous solution containing 5 mM of [(NH₄)₂MoS₄] and 0.5 M Na₂SO₄ under 1 sun illumination for 1,200 s. After film preparation, the electrode was removed from the prepared media and washed with 100 ml distilled water. Then the electrode was disassembled from the epoxy cover and annealed in Ar atmosphere for 4 h at 450 °C. A similar procedure as previously described was applied to reassemble the annealed GaInP₂ electrode.

IPCE measurement. Incident photon-to-current efficiency (IPCE) was performed in a three-electrode configuration, with Ag/AgCl as the reference electrode and a Pt foil counter electrode, in 0.5 M H₂SO₄ aqueous solution (pH 0.3) electrolyte at -0.2 V versus Ag/AgCl (0 V versus RHE). The set-up of the instrument was reported before³⁹. The sequence at each wavelength was 6 s of dark, following 4 s of illumination. The current was collected at 10 points per second, with the final 10 points of each light and dark cycle averaged. For each data point at one wavelength, the photocurrent was obtained by subtracting the dark current from the light current. Sample photocurrent data were normalized to the output of a calibrated silicon photodiode (Hamamatsu, A02 S1336-8BQB) outside the solution. Each plot was obtained by the average value of two experiments.

X-ray photoelectron spectroscopy measurement. Photoelectron spectroscopy was performed on a Kratos AXIS Nova or Physical Electronics (PHI) 5600 system depending on instrument availability. In either case, photoelectrons were generated using monochromatic Al K α X-rays at 1,486.6 eV. Base pressures were better than 1×10^{-9} torr. Binding energy calibrations were performed by comparing measured Au 4f 7/2, Ag 3d 5/2 and Cu 2p 3/2 core level spectra from clean metal foils and the accepted centroid positions reported by Seah⁴⁰. For E3 samples that exhibited slight charging during measurement, the C 1s peak was referenced to 284.6 eV (ref. 41). Least-squares fitting of XPS data was done using CasaXPS software with combinations of Gaussian (80%) and Lorentzian (20%) peak shapes. Constraints on the Mo 3d_{5/2} and 3d_{3/2} peaks included peak splitting of 3.18 eV and an intensity ratio of 3:2 respectively. Constraints on the S 2p_{3/2} and 2p_{1/2} peaks included peak splitting of 1.20 eV and an intensity ratio of 2:1 respectively. Calculations were performed using the TPP-2 formula for electron effective attenuation lengths (EALs). The resulting thicknesses assumed a dense MoS₂ coating and used the National Institute of Standards and Technology (NIST) EAL database with inputs: MoS₂ density, 5.06 g cm⁻³; asymmetry (beta parameter), 1.18 averaged between Mo 3d_{5/2} and S 2p_{3/2} orbitals; KE, 1,300 eV; valence, 18; $E_g = 1.23$ eV and using our XPS set-up geometry. These calculations indicate that the photoelectron signal attenuation reaches 99% at a film thickness of 10.0 nm. For a MoS₂ thickness of 6.5 nm the signal is attenuated by 95%. Hence, with a calculated effective attenuation length of 2.193 nm, and very good fits to the XPS data, the analysis presented represents the top 10 nm of the coating.

Scanning electron microscope measurement. The surface morphology of the TiO₂-modified GaInP₂ electrode or Pt-catalyst-modified GaInP₂ was analysed using a JEOL JSM 7600F field-emission scanning electron microscope operated at 5 kV. Characterization of the GaInP₂-TiO₂-MoS₂/MoO₃ surface before and after annealing indicates the formation of more-defined surface grains, with sizes of

100–200 nm after annealing, as shown by SEM measurements (Supplementary Fig. 5).

Atomic layer deposition of TiO₂. The TiO₂ was deposited onto the GaInP₂ following a published procedure. TiO₂ (24–30 nm) was deposited onto the GaInP₂ surface to protect the surface from corrosion. The thickness of the TiO₂ layer was characterized by STEM.

Sputtering of platinum and ruthenium. The Pt/Ru sputtering is performed following a literature procedure⁴². The samples were mounted to a wheel inside the sputtering chamber. The wheel serves two purposes: to move the samples out of the way (to the top of the wheel) while the sputter head is being warmed up or powered down; and to pass the samples through the sputtering plume at a defined rate. With samples mounted, the chamber is evacuated to $\sim 1 \times 10^{-6}$ torr base pressure before introducing and maintaining a 10 mtorr argon background. With samples in the 'up' position, the sputter head power supply is adjusted to 20 W d.c., establishing an Ar plasma, which is allowed to stabilize for 2 min. Then, Pt/Ru sputtering proceeds as two wheel rotations at 15 r.p.m. so that the samples pass underneath the sputter head twice.

Determination of the turnover number and turnover frequency. After the catalyst loading amount was determined by ICP-MS, the turnover number was calculated by the following equation: $= (C/2Fn)$, where C is the Coulomb passed during the electrolysis process. F is the Faraday constant in C mol⁻¹. n is the mole amount of catalyst deposited on top of the electrode surface. TOF is determined by TON per second.

Data availability. The data that support the findings of this study are available from the corresponding author (J.G.) on request.

Received 17 May 2016; accepted 17 November 2016;
published 9 January 2017

References

- Haryanto, A., Fernando, S., Murali, N. & Adhikari, S. Current status of hydrogen production techniques by steam reforming of ethanol: a review. *Energy Fuels* **19**, 2098–2106 (2005).
- Barelli, L., Bidini, G., Gallorini, F. & Servili, S. Hydrogen production through sorption-enhanced steam methane reforming and membrane technology: a review. *Energy* **33**, 554–570 (2008).
- Walter, M. G. *et al.* Solar water splitting cells. *Chem. Rev.* **110**, 6446–6473 (2010).
- Lewis, N. S. Light work with water. *Nature* **414**, 589–590 (2001).
- Iwashina, K. & Kudo, A. Rh-doped SrTiO₃ photocatalyst electrode showing cathodic photocurrent for water splitting under visible-light irradiation. *J. Am. Chem. Soc.* **133**, 13272–13275 (2011).
- Zhen, C. *et al.* Nonstoichiometric rutile TiO₂ photoelectrodes for improved photoelectrochemical water splitting. *Chem. Commun.* **49**, 6191–6193 (2013).
- Khaselev, O. & Turner, J. A. A monolithic photovoltaic-photoelectrochemical device for hydrogen production via water splitting. *Science* **280**, 425–427 (1998).
- Kocha, S. S., Montgomery, D., Peterson, M. W. & Turner, J. A. Photoelectrochemical decomposition of water utilizing monolithic tandem cells. *Solar Energy Mater. Solar Cells* **52**, 389–397 (1998).
- Döscher, H., Young, J., Geisz, J., Turner, J. & Deutsch, T. Solar-to-hydrogen efficiency: shining light on photoelectrochemical device performance. *Energy Environ. Sci.* **9**, 74–80 (2016).
- Khaselev, O. & Turner, J. A. Electrochemical stability of p-GaInP₂ in aqueous electrolytes toward photoelectrochemical water splitting. *J. Electrochem. Soc.* **145**, 3335–3339 (1998).
- Wang, H., Deutsch, T., Welch, A. & Turner, J. A. The stability of illuminated p-GaInP₂ semiconductor photoelectrode. *Int. J. Hydrog. Energy* **37**, 14009–14014 (2012).
- Hu, S. *et al.* Amorphous TiO₂ coatings stabilize Si, GaAs, and GaP photoanodes for efficient water oxidation. *Science* **344**, 1005–1009 (2014).
- Liu, M., Nam, C.-Y., Black, C. T., Kamcev, J. & Zhang, L. Enhancing water splitting activity and chemical stability of zinc oxide nanowire photoanodes with ultrathin titania shells. *J. Phys. Chem. C* **117**, 13396–13402 (2013).
- Gu, J. *et al.* Water reduction by a p-GaInP₂ photoelectrode stabilized by an amorphous TiO₂ coating and a molecular cobalt catalyst. *Nat. Mater.* **15**, 456–460 (2015).
- Lee, M. H. *et al.* p-type InP nanopillar photocathodes for efficient solar-driven hydrogen production. *Angew. Chem.* **124**, 10918–10922 (2012).
- Krstajić, N. *et al.* Electrodeposition of Ni–Mo alloy coatings and their characterization as cathodes for hydrogen evolution in sodium hydroxide solution. *Int. J. Hydrog. Energy* **33**, 3676–3687 (2008).

17. Tributsch, H. & Bennett, J. Electrochemistry and photochemistry of MoS₂ layer crystals. I. *J. Electroanal. Chem. Interface Electrochem.* **81**, 97–111 (1977).
18. Yan, Y., Xia, B., Xu, Z. & Wang, X. Recent development of molybdenum sulfides as advanced electrocatalysts for hydrogen evolution reaction. *ACS Catal.* **4**, 1693–1705 (2014).
19. Tran, P. D. *et al.* Novel assembly of an MoS₂ electrocatalyst onto a silicon nanowire array electrode to construct a photocathode composed of elements abundant on the Earth for hydrogen generation. *Chem.-A Eur. J.* **18**, 13994–13999 (2012).
20. Morales-Guio, C. G., Tilley, S. D., Vrubel, H., Grätzel, M. & Hu, X. Hydrogen evolution from a copper (I) oxide photocathode coated with an amorphous molybdenum sulphide catalyst. *Nat. Commun.* **5**, 3059 (2014).
21. Jin, J. *et al.* An experimental and modeling/simulation-based evaluation of the efficiency and operational performance characteristics of an integrated, membrane-free, neutral pH solar-driven water-splitting system. *Energy Environ. Sci.* **7**, 3371–3380 (2014).
22. Wang, T. & Gong, J. Single-crystal semiconductors with narrow band gaps for solar water splitting. *Angew. Chem. Int. Ed.* **54**, 10718–10732 (2015).
23. Kocha, S. S. & Turner, J. A. Displacement of the bandedges of GaInP₂ in aqueous electrolytes induced by surface modification. *J. Electrochem. Soc.* **142**, 2625–2630 (1995).
24. Merki, D., Fierro, S., Vrubel, H. & Hu, X. Amorphous molybdenum sulfide films as catalysts for electrochemical hydrogen production in water. *Chem. Sci.* **2**, 1262–1267 (2011).
25. Ponomarev, E., Neumann-Spallart, M., Hodes, G. & Levy-Clement, C. Electrochemical deposition of MoS₂ thin films by reduction of tetrathiomolybdate. *Thin Solid Films* **280**, 86–89 (1996).
26. Deutsch, T. *et al.* Stable photoelectrode surfaces and methods. US patent US20140332374 A1 (2014).
27. McKone, J. R., Pieterick, A. P., Gray, H. B. & Lewis, N. S. Hydrogen evolution from Pt/Ru-coated p-type WSe₂ photocathodes. *J. Am. Chem. Soc.* **135**, 223–231 (2012).
28. Seger, B. *et al.* Using TiO₂ as a conductive protective layer for photocathodic H₂ evolution. *J. Am. Chem. Soc.* **135**, 1057–1064 (2013).
29. Pourbaix, M. *Atlas of Electrochemical Equilibria in Aqueous Solutions* (Pergamon, 1966).
30. Bard, A. J., Faulkner, L. R., Leddy, J. & Zoski, C. G. *Electrochemical Methods: Fundamentals and Applications* Vol. 2 (Wiley, 1980).
31. Benck, J. D., Chen, Z., Kuritzky, L. Y., Forman, A. J. & Jaramillo, T. F. Amorphous molybdenum sulfide catalysts for electrochemical hydrogen production: insights into the origin of their catalytic activity. *ACS Catal.* **2**, 1916–1923 (2012).
32. Vrubel, H. & Hu, X. Growth and activation of an amorphous molybdenum sulfide hydrogen evolving catalyst. *ACS Catal.* **3**, 2002–2011 (2013).
33. Benoist, L. *et al.* X-ray photoelectron spectroscopy characterization of amorphous molybdenum oxysulfide thin films. *Thin Solid Films* **258**, 110–114 (1995).
34. Benck, J. D., Hellstern, T. R., Kibsgaard, J., Chakthranont, P. & Jaramillo, T. F. Catalyzing the hydrogen evolution reaction (HER) with molybdenum sulfide nanomaterials. *ACS Catal.* **4**, 3957–3971 (2014).
35. Zhang, L., Liu, C., Wong, A. B., Resasco, J. & Yang, P. MoS₂-wrapped silicon nanowires for photoelectrochemical water reduction. *Nano Res.* **8**, 281–287 (2015).
36. Chen, Z. *et al.* Core-shell MoO₃-MoS₂ nanowires for hydrogen evolution: a functional design for electrocatalytic materials. *Nano Lett.* **11**, 4168–4175 (2011).
37. Hu, X., Chen, L. & Chen, Y. Universal and versatile MoO₃-based hole transport layers for efficient and stable polymer solar cells. *J. Phys. Chem. C* **118**, 9930–9938 (2014).
38. Mohamed, M. M. Effect of ceria-doped titania on the structure and acidic properties of MoO₃/TiO₂ catalysts. *Appl. Catal. A* **267**, 135–142 (2004).
39. Gu, J. Y. Y., Young, J. L., Steirer, K., Neale, N. R. & Turner, J. A. Water reduction by a p-GaInP₂ photoelectrode stabilized by an amorphous TiO₂ coating and a molecular cobalt catalyst. *Nat. Mater.* **15**, 456–460 (2015).
40. Seah, M. Post-1989 calibration energies for X-ray photoelectron spectrometers and the 1990 Josephson constant. *Surf. Interface Anal.* **14**, 488 (1989).
41. Tsang, S. C., Caps, V., Paraskevas, I., Chadwick, D. & Thompson, D. Magnetically separable, carbon-supported nanocatalysts for the manufacture of fine chemicals. *Angew. Chem.* **116**, 5763–5767 (2004).
42. Dameron, A. A. *et al.* Pt–Ru alloyed fuel cell catalysts sputtered from a single alloyed target. *ACS Catal.* **1**, 1307–1315 (2011).

Acknowledgements

This material is based on work supported by the US Department of Energy, Office of Science, Office of Basic Energy Sciences, Solar Photochemistry Program under contract number DEAC36-08GO28308. We gratefully acknowledge C. Antunes for ICP-MS measurement, A. Norman for the helpful discussions and plan-view TEM measurement for the PtRu–GaInP₂ electrode.

Author contributions

J.G., J.A.A., S.F., N.R.N. and J.A.T. wrote the manuscript. J.G. and S.F. conceived the experiments. J.G. conducted the photoelectrochemical characterization. J.A.A. conducted the STEM, EDS and EELS measurements. J.A.A. and M.A.-J. analysed and interpreted the STEM, EDS and EELS data. K.X.S. conducted the XPS experiment and related analysis. Y.Y. conducted the initial testing for MoS_x catalyst deposition. C.X. conducted the SEM measurements and J.L.Y. conducted the atomic layer TiO₂ deposition.

Additional information

Supplementary information is available for this paper.

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How to cite this article: Gu, J. *et al.* A graded catalytic-protective layer for an efficient and stable water-splitting photocathode. *Nat. Energy* **2**, 16192 (2017).

Competing interests

The authors declare no competing financial interests.